EE284A Project Proposal

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Part 1: System Design

Application domain: Smart home IoT (shared living environments)

Objective
Dorm living involves shared spaces with limited visibility into roommate status (e.g., whether someone is inside, asleep, or hosting guests). This often leads to inconveniences such as interrupting sleep, lack of privacy, or inefficient coordination between roommates.

Also, students spend significant time indoors with irregular schedules, making awareness of room usage and conditions valuable for both comfort and productivity.

Personally, we have experienced situations where lack of communication causes awkward encounters with our roommates, so we are motivated to create a system that would resolve this.

Value Proposition

- ➔ Convenience: Instantly know room status without texting
- ➔ Privacy-preserving: No cameras, only abstracted sensor data
- ➔ Low-cost: <$30 additional hardware
- ➔ Non-intrusive: Runs in background without user input
- ➔ Long term impact: scalable to apartments and other shared living spaces

Initial thoughts

- ➔ *Functions*
  - designed for 2 roommates
  - Sends roommate status via iPhone widget
  - Detects presence of people in the room (and number of people)
  - Senses whether resident is asleep
  - Transmit real-time status to cloud
  - Display status via mobile widget
- ➔ *Data*
  - Distance / movement patterns (ultrasonic sensors)
  - Door open / close events (magnetic switch)
  - Ambient light levels with a light sensor

- *Privacy and Security*
  - No cameras or audio recording
  - No personally identifiable data stored
- *Reliability and Maintenance*
  - Multi-sensor fusion improves robustness
  - Door sensor + motion reduces counting errors

Part 2: Hardware Proof-Of-Concept

Features

- Sends roommate status via iPhone widget
- Detects presence of people in the room
- Estimates the number of people in the room
- Senses whether resident is asleep
- Transmit real-time status to cloud
- Display status via mobile widget

Inputs/Outputs

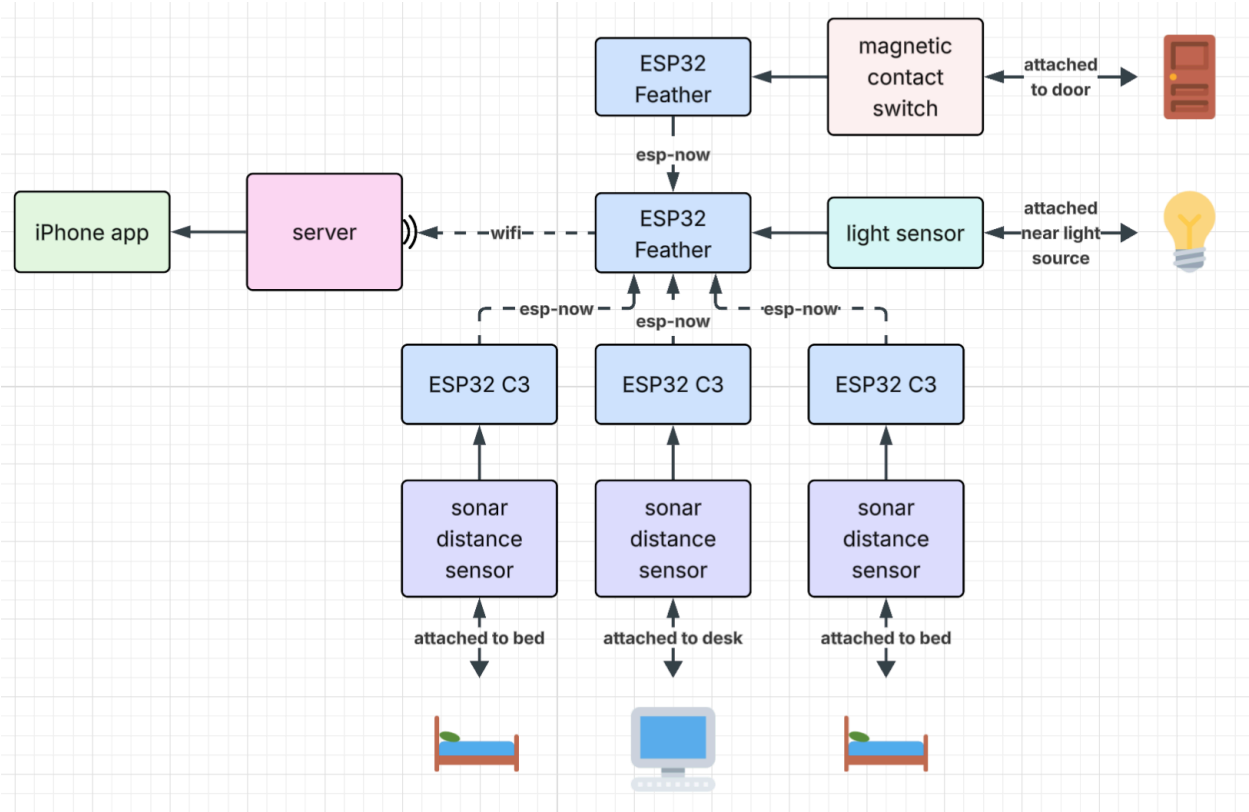
Inputs:

- Presence of people in the room
- Presence of resident in bed
- Opening the door
- Light in the room on/off

Outputs:

- Status of the room in an iPhone widget available to both roommates

Block Diagram



BOM

Part Number	Cost (USD)	Link	Quantity
Magnetic contact switch (door sensor)	3.75	Magnetic Door Switch Set - SparkFun Electronics	1
Analog Light Sensor	1.75	Ambient Light Sensor - TEMT6000x01 - SparkFun Electronics	1
RCWL-1601, HC-SR04 compatible Ultrasonic Sonar Distance Sensor	10.77 for 3	HC-SR04 Ultrasonic Sonar Distance Sensor	3
Seeed Studio ESP32-C3 (3PCS)	13.49	102010572 Seeed Technology Co., Ltd | RF, RFID, Wireless Evaluation Boards |	1

		DigiKey	
Total Cost	**29.76**		

Technical Innovation Requirement

This project integrates sensor fusion to parse motion, light, and door activity inputs in order to infer roommate occupancy and sleep state.

We will use wall outlets with USB-C connectors, which provide 5V.

We will also be sending data to an iPhone and have a unique visualization widget.